MTH2003 Examples Sheet 2

Learn how to solve second order ODEs by practising all of these examples. Feel free to work on these examples with your friends and ask me about any queries during the drop-in Q/A sessions on Mondays at 12:35 and 16:35 (see your timetable)

1. For each of the following differential equations for $y(t)$:

   [i] Find the complementary solution.
   [ii] Find a particular solution.
   [iii] Write down the general solution.
   [iv] Where the initial conditions are given, find also the solution satisfying these initial conditions.

   ($\boldsymbol{a}$) $y'' + 8y' + 15y = -3\,\mathrm{e}^{-4t}$, $y(0) = 6$, $y'(0) = -25$.
   ($\boldsymbol{b}$) $y'' + 6y' + 8y = 16\,\mathrm{e}^{-2t}$.
   ($\boldsymbol{c}$) $y'' + 4y' + 13y = 136\cos(5t)$, $y(0) = 0$, $y'(0) = 25$.
   ($\boldsymbol{d}$) $y'' + 8y' + 25y = 125t^2 + 5t - 14$, $y(0) = 5$, $y'(0) = -11$.
   ($\boldsymbol{e}$) $y'' + 6y' + 8y = \mathrm{e}^{3t}\sin(2t)$.
   ($\boldsymbol{f}$) $y'' - 5y' + 6y = 4t\,\mathrm{e}^{4t}$, $y(0) = 5$, $y'(0) = 11$.

2. For each of the following homogeneous equations:

   [i] Show that the given function $y_1(t)$ is a solution of that equation.
   [ii] Using the Method of Reduction of Order, substitute $y(t) = v(t)y_1(t)$, where $v(t)$ is a new unknown function, into the differential equation. Obtain an equation for v' and hence find the general solution for v, the general solution for y, and from there identify the second linearly independent solution $y_2(t)$.
   [iii] Substitute this expression for y_2 into the differential equation to check that it is a solution.
   [iv] Verify, by calculating the Wronskian of $y_1(t)$ and $y_2(t)$, that solution $y_2(t)$ is linearly independent from $y_1(t)$.

   ($\boldsymbol{a}$) $t(t-2)\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + (2-t^2)\dfrac{\mathrm{d}y}{\mathrm{d}t} + 2(t-1)y = 0, \quad t \neq 0, 2, \quad y_1 = \mathrm{e}^t.$
   ($\boldsymbol{b}$) $2t^2\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} + 3t\dfrac{\mathrm{d}y}{\mathrm{d}t} - y = 0, \quad t > 0, \quad y_1 = 1/t.$
   ($\boldsymbol{c}$) $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2} - 2\dfrac{\mathrm{d}y}{\mathrm{d}t} + y = 0, \quad y_1 = \mathrm{e}^t.$
   ($\boldsymbol{d}$) $t^2y'' - 4ty' + 6y = 0, \quad t \neq 0, \quad y_1 = t^2.$

3. Solve each of the following second-order linear nonhomogeneous equations using the Method of Variation of Parameters.

   ($\boldsymbol{a}$) $y'' + 6y' + 9y = \frac{1}{t^2}\,e^{-3t}, \quad t \neq 0.$
   ($\boldsymbol{b}$) $2t^2y'' + 3ty' - y = t^2\ln t, \quad t > 0\ .$

4. Solve the Euler-Cauchy equations:

   ($\boldsymbol{a}$)
   $$t^2y'' - 3ty' + 4y = 0, \quad t > 0\,.$$
   ($\boldsymbol{b}$)
   $$t^2y'' + 3ty' + 5y = 0, \quad t > 0\,.$$